

Quiz

Question	Options	Answer
What is the result of the following expression in Java? int result = 10 + 5 * 2;	20 30 15 25	20 Correct! In Java, multiplication has a higher precedence than addition. So, 5 * 2 is evaluated first to get 10, and then 10 + 10 equals 20.
int x = 5; x += 3;	3 5 8 15	8 Correct! The += is a shorthand assignment operator. x += 3; is exactly the same as writing x = x + 3;. Since x was 5, 5 + 3 gives 8.
Which of the following comparison expressions evaluates to true in Java?	5 > 10 15 == 10 20 <= 20 8 != 8	20 <= 20 Correct! The <= operator means "less than or equal to". Since 20 is equal to 20, the expression evaluates to true.
Assume int a = 10; and int b = 5; . What does the following expression evaluate to? !(a > b) && (b < 20)	True False Null It will cause compilation error	False Correct! First, a > b evaluates to true. Applying the NOT operator! turns it to false. Next, b < 20 is true. Finally, false && true evaluates to false.
What is the result of the following Java expression? boolean b = (5 % 2 == 1) (3 * 2 == 5);	True False This expression throws an Arithmetic exception 0	true Correct! 5 % 2 gives a remainder of 1, so 1 == 1 is true. Because of the short-circuit OR (), if the first part is true, Java knows the whole expression is true without evaluating the second part.
Modulus & Division Mix int result = 15 - 9 % 4 * 2 + 6 / 3; System.out.println(result);	• What prints out? • Hint: Modulus %, multiplication *, and division / all have the same precedence level. Use left-to-right associativity for them before handling addition and subtraction.	
The Logical Operator int x = 5; boolean check = (x++ > 5) && (++x < 10) (x == 6); System.out.println("check = " + check + ", x = " + x);	• What prints out for check and x? • Hint: Look out for "short-circuit" evaluation. If the first part of an && (AND) statement is false, Java won't even look at the second part! Also, remember the difference between postfix x++ (use then change) and prefix ++x (change then use).	
String Concatenation System.out.println("Result: " + 5 + 10 * 2);	• What text prints to the console? • Hint: The + operator acts as mathematical addition <i>unless</i> one of the sides is a String, which turns it into text gluing (concatenation).	